

# Jesús Ramírez Nava

Senior Software Engineer · Monterrey, Mexico

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Senior software engineer with 10 years in C/C++ — from 16-bit microcontrollers and CAN bus tooling at John Deere to Amazon Fire TV middleware and API backends today. I've led teams of 10+ engineers, served as Scrum Master, and designed the simulators, test tooling, and backends other teams depend on. Independently, I build web and AI products for clients. Based in Monterrey (UTC-6) and open to remote work with US teams.

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## EXPERIENCE

### Independent Software Engineer · [jesusrtech](#)

Jun 2026 – Present

*Project: carterapp — offline-first personal finance app (carterapp.com.mx)*

- \_ Designed and built carterapp, a personal finance PWA for Mexico: log an expense in three taps, in Spanish or English, with MXN and USD kept side by side.
- \_ Designed an offline-first sync model — records are written to the device first and synced with device-minted IDs, so entry never waits for the network and sync is idempotent.
- \_ Enforced data integrity in the database itself: composite foreign keys keep every transaction in its account's currency and its category's type.
- \_ Built monthly budgets, a card cut-off and payment calendar, a month-end balance projection, and the backend for subscriptions and push notifications.

React   TypeScript   PWA   IndexedDB (Dexie)   Supabase   PostgreSQL   Stripe   Vercel

### Independent Software Engineer · [jesusrtech](#)

Jun 2026 – Present

*Project: Mi 70-20-10 — leadership coaching platform for eQual Business Solutions*

- \_ Designed and built a coaching platform where participants, coaches, and admins run a leadership-development program end to end.
- \_ Secured the data with Postgres row-level security per role, so each user sees exactly what their role allows and roles can't be changed from the client.
- \_ Built AI-assisted rubric evaluation of submissions, an AI coach chat, and retrieval over the program's material, with coaches approving what is released.
- \_ Extracted a reusable React component library and automated deploys for the app and its backend functions.

React   TypeScript   Supabase   PostgreSQL (RLS)   Edge Functions   Claude API  
Gemini API   Vitest   Vercel

### Senior Software Engineer · [Svitla Systems](#)

Mar 2026 – Present

*Project: API backend abstraction to migrate a legal reporting system to a newer platform version*

- \_ Designing C++ features for an API backend abstraction layer that lets a legal reporting system migrate to its new platform version without breaking existing clients.
- \_ Defining clean interfaces between the legacy C++ core and the new API so features can move over incrementally.
- \_ Implementing new features and fixing defects through root cause analysis, working with beta users and QA.
- \_ Integrating AI-assisted features into the backend and tuning them for response time and cost.

C++   API design   Software architecture   React   AI integration

## Senior Software Engineer · GlobalLogic

Aug 2023 – Dec 2025

*Project: Amazon Fire TV next generation — porting a third-party video streaming app to new devices and Amazon's new non-Android OS*

- \_ Ported a major third-party video streaming app to Amazon's next-generation Fire TV devices, working in the middleware layer.
- \_ Adapted Google's Cobalt application runtime to the new device hardware and to Amazon's new operating system, which is not based on Android.
- \_ Implemented features and fixed defects through root cause analysis, working with beta users and test teams.
- \_ Coordinated with partner engineering teams to route and escalate issues outside the project's scope.

C++ Google Cobalt Middleware API/ABI integration Third-party SDK integration Jira

## Senior Software Engineer · GlobalLogic

Aug 2023 – Dec 2025

*Project: Test environment to automate embedded-system release processes for an engineering-services provider*

- \_ Refactored a key desktop application with Python and PySimpleGUI, significantly improving readability and maintainability.
- \_ Strengthened the architecture by applying SOLID principles and the listener pattern, producing a more robust, understandable codebase.
- \_ Designed and built a project-manager feature from scratch to streamline creating and editing test profiles.

Python PySimpleGUI Git C++ Visual C++ Jira

## Independent Software Engineer · jesusrtech

Jan 2023 – Jun 2023

*Project: Antártico — order management for cold-storage spaces (antartico.com.mx)*

- \_ Designed and built a web app to manage orders for cold-storage spaces, from the data model to deployment, in about six months.
- \_ Built it on Laravel and SQL behind a login, and deployed and ran it on a DigitalOcean server.

Laravel PHP SQL DigitalOcean Nginx

## Software Tech Lead · 3Pillar Global

Apr 2022 – Aug 2023

- \_ Analyzed and fixed defects in existing features written in C++ and WinCC OA.
- \_ Developed new functionality and unit tests, including keeping system status in sync with the data-queue handshake.
- \_ Designed and implemented DevOps infrastructure to support a new licensing model, enabling migration to a modern tech stack.
- \_ Built from scratch a Python 3 proof-of-concept API wrapper over crypto.com to simplify requests from higher abstraction layers — the bid was accepted.
- \_ Leveled up in Go, Python, API design, and systems design through company training.

C++ WinCC OA Go Python AWS Docker CI/CD Linux Git

## Software Supervisor / Technical Lead · John Deere

Feb 2021 – Mar 2022

- \_ Led the Mexico team as tech lead and developer, fixing bugs in C++ engine-testing applications.
- \_ Supervised 10+ engineers across 5 clients, guiding their professional and personal career paths.
- \_ Interviewed candidates and designed team processes to onboard new engineers during headcount growth.

C C++ Qt RTOS Python CI/CD Linux Git

## Senior Embedded Software Engineer · John Deere

Dec 2017 – Jan 2021

- \_ Scrum Master and developer for embedded applications on 16-bit microcontrollers (C) and their GUIs (Qt C++).
- \_ Led the Mexico Qt C++ team troubleshooting the tractor UI — from microcontroller buffer logic to data interpretation and on-screen visualization.
- \_ Stood up and led a new team building simulators for the 16-bit microcontroller, the CAN bus, and virtual product interactions in C and C#.

- \_ Led design and implementation of a virtual Ethernet simulation supporting Ethernet on 32-bit FreeRTOS microcontrollers.
- \_ Reported to users and stakeholders every 2-week sprint and presented the plan for each upcoming PSI.

C C++ C# Qt FreeRTOS CAN Ethernet Python Docker CI/CD Linux

### Embedded Software Engineer · John Deere

Dec 2016 – Dec 2017

- \_ Developed Qt UI applications, frontend and backend, in C++.
- \_ Designed features to process data from a 16-bit microcontroller over CAN (J1939).
- \_ Improved code quality by fixing inconsistencies and writing unit tests with Google Test.

C++ Qt CAN J1939 Google Test Jenkins Mercurial Docker Linux

### Embedded Software Intern · John Deere

May 2016 – Dec 2016

- \_ Designed and built from scratch a static code analyzer with a GUI that highlights bad coding practices, using Flex and Bison.
- \_ Built electronic test boxes that emulate signals and communication protocols for device testing.

C C++ CAN Flex Bison HTML CSS

### Computer Science Assistant · Tec de Monterrey

Jan 2015 – Aug 2015

- \_ Wrote the base code for Kinect applications to improve the classroom experience through technology.
- \_ Built Kinect interactions with hardware interfaces and controlled XAML apps through natural body movement in C#.

C# Kinect SDK XAML

## EDUCATION

### B.Sc. Digital Systems and Robotics Engineering

Tecnológico de Monterrey, Campus Monterrey

2016  
MONTERREY, MEXICO

### Technical Degree in Computer Science

ICE Computación

2011

### Technical Degree in Computer Science / Business Administration

CBTis 67

2010

## SKILLS

### LANGUAGES

C C++ Python C# Go SQL PHP

### EMBEDDED

16/32-bit microcontrollers FreeRTOS CAN / J1939 Ethernet Hardware simulation

### FRAMEWORKS & UI

Qt React Laravel PySimpleGUI Google Test

### BACKEND & SYSTEMS

API design API/ABI integration Third-party SDKs AI backends Systems design

### DEVOPS & TOOLS

Docker AWS Jenkins CI/CD Linux Git Jira

### LEADERSHIP

Tech lead Scrum Master Team of 10+ Hiring & onboarding

PROJECTS & AWARDS

|             |                                                                                                                                                                   |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2019 - 2024 | Designed and built a prototype of a next-generation learning management system (LMS) with Laravel and SQL, hosted on DigitalOcean.                                |
| 2020 - 2021 | Designed and built a web app that manages orders for cold-storage spaces, using Laravel and SQL on DigitalOcean.                                                  |
| 2016        | Built a MISRA-C analyzer that checks C code against the safety-critical coding standard, using Flex and Bison for syntax and semantic analysis.                   |
| 2015        | Built an open-source Android app for programming a robot over Bluetooth with interactive, graphical instructions.                                                 |
| 2015        | Built a computer-vision system for a Parrot drone that recognized objects around it and responded with matching flight routines.                                  |
| 2015        | Parallelized a file-decryption program in C to speed it up.                                                                                                       |
| 2015        | Created Robython, a high-level programming language and translator that lets children program hardware.                                                           |
| 2014        | Built a complete data-acquisition system for electrical conductivity: firmware in C plus a C# desktop app that reads the microprocessor and shares data over FTP. |
| 2014        | Built an IoT platform that monitors biosignals in real time with humidity, temperature, and heart-rate sensors.<br>Winner, Intel challenge – SISCTI 40            |
| 2010        | Designed an automatic, programmable pet-food dispenser in C.<br>State winner, DGETI prototypes competition – national finalist                                    |

LANGUAGES

Spanish — Native      English — B2

CERTIFICATIONS

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|-------------------------------------|-------------------------|
| Generative AI Leader                | GOOGLE CLOUD · AUG 2025 |
| Google Cybersecurity Specialization | GOOGLE · DEC 2023       |